

# week\_8\_lab\_airbnb

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Week 8 instruction for Airbnb file: All\_chch listings (separate datasets from oct 2025 to jun 2026 combined, but only for chch city). clean the dataset -> ultimate aim is going to be comparing rental prices and number of available properties with tenancy dataset

```
##          id          name      host_id      host_name
## Min.      :2.040e+05  Length   :28795  Min.      :5.738e+04  Length   :28795
## 1st Qu.   :5.237e+07  N.unique : 4854  1st Qu.   :6.396e+07  N.unique : 1422
## Median    :1.090e+18  N.blank  :    0  Median    :1.721e+08  N.blank  :    0
## Mean      :9.031e+17  Min.nchar:    3  Mean      :1.988e+15  Min.nchar:    1
## 3rd Qu.   :1.394e+18  Max.nchar:   68  3rd Qu.   :4.471e+08  Max.nchar:   40
## Max.      :1.710e+18                      Max.      :1.705e+18  NAs       :    1
##
## neighbourhood_group  neighbourhood  latitude  longitude
## Length      :28795  Length   :28795  Min.      : -43.87  Min.      :172.4
## N.unique     :    1  N.unique   :   16  1st Qu.   : -43.56  1st Qu.   :172.6
## N.blank      :    0  N.blank    :    0  Median    : -43.53  Median    :172.6
## Min.nchar    :   17  Min.nchar   :   10  Mean      : -43.56  Mean      :172.7
## Max.nchar    :   17  Max.nchar   :   20  3rd Qu.   : -43.52  3rd Qu.   :172.7
##                      Max.      : -43.42  Max.      :173.1
##
##          room_type      price      minimum_nights  number_of_reviews
## Length      :28795  Min.      :   27.0  Min.      :   1.000  Min.      :   0.00
## N.unique     :    4  1st Qu.   :  140.0  1st Qu.   :   1.000  1st Qu.   :   5.00
## N.blank      :    0  Median    :  205.0  Median    :   1.000  Median    :  26.00
## Min.nchar    :   10  Mean      :  260.2  Mean      :   1.788  Mean      :  70.39
## Max.nchar    :   15  3rd Qu.   :  284.0  3rd Qu.   :   2.000  3rd Qu.   :  85.00
##                      Max.      :43654.0  Max.      :365.000  Max.      :1584.00
##                      NAs       :10667  NAs       :37
## last_review      reviews_per_month  calculated_host_listings_count
## Min.      :2014-04-14  Min.      : 0.010  Min.      :   1.0
## 1st Qu.   :2025-10-05  1st Qu.   : 0.760  1st Qu.   :   1.0
## Median    :2026-01-13  Median    : 1.700  Median    :   3.0
## Mean      :2025-11-05  Mean      : 2.351  Mean      :  58.3
## 3rd Qu.   :2026-03-20  3rd Qu.   : 3.330  3rd Qu.   :  47.0
## Max.      :2026-06-21  Max.      :16.570  Max.      :1586.0
## NAs       :2627      NAs       :2627
## availability_365  number_of_reviews_ltm  license      month_year
## Min.      :   0.0  Min.      :    0  Mode:logical  Length   :28795
## 1st Qu.   :109.0  1st Qu.   :    2  NAs :28795  N.unique :    9
## Median    :225.0  Median    :   10  N.blank  :    0
## Mean      :203.8  Mean      :   19  Min.nchar :    7
## 3rd Qu.   :302.0  3rd Qu.   :   27  Max.nchar :    7
```

```
## Max.      :365.0      Max.      :185
##
```

data has: Rows: 28795 Columns: 19

```
## Rows: 28,795
## Columns: 19
## $ id          <dbl> 4.018377e+07, 1.149729e+18, 1.097487e+0~
## $ name        <chr> "Ensuite Cabin (Sleeps 6)", "Handy city~
## $ host_id     <dbl> 98758734, 298456605, 4993285, 54299306,~
## $ host_name   <chr> "North South Holiday Park", "Tony", "Pa~
## $ neighbourhood_group <chr> "Christchurch City", "Christchurch City~
## $ neighbourhood <chr> "Harewood Ward", "Central Ward", "Fenda~
## $ latitude    <dbl> -43.47002, -43.53270, -43.51149, -43.48~
## $ longitude   <dbl> 172.5709, 172.6640, 172.6254, 172.6687,~
## $ room_type   <chr> "Private room", "Private room", "Privat~
## $ price       <dbl> NA, NA, 166, NA, 126, 585, 145, 450, NA~
## $ minimum_nights <dbl> 1, 1, 1, 2, 1, 2, 1, 2, 1, 1, 7, 1, 1, ~
## $ number_of_reviews <dbl> 0, 25, 230, 2, 8, 0, 6, 0, 1, 0, 1, 5, ~
## $ last_review <date> NA, 2025-07-15, 2025-03-03, 2025-04-26~
## $ reviews_per_month <dbl> NA, 1.47, 2.17, 0.35, 0.58, NA, 0.26, N~
## $ calculated_host_listings_count <dbl> 20, 6, 1, 1, 20, 3, 4, 1, 2, 6, 4, 14, ~
## $ availability_365 <dbl> 317, 0, 221, 0, 359, 78, 348, 363, 4, 0~
## $ number_of_reviews_ltm <dbl> 0, 21, 14, 2, 5, 0, 2, 0, 0, 0, 1, 4, 2~
## $ license     <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ month_year  <chr> "2025-10", "2025-10", "2025-10", "2025-~
```

```
## # A tibble: 28,795 x 19
```

```
##       id name      host_id host_name neighbourhood_group neighbourhood latitude
##       <dbl> <chr>      <dbl> <chr>      <chr>              <chr>          <dbl>
## 1 4.02e 7 Ensuite~ 9.88e7 North So~ Christchurch City Harewood Ward -43.5
## 2 1.15e18 Handy c~ 2.98e8 Tony      Christchurch City Central Ward -43.5
## 3 1.10e 6 Heritag~ 4.99e6 Paulette  Christchurch City Fendalton Wa~ -43.5
## 4 1.34e18 Settle ~ 5.43e7 Sarena     Christchurch City Burwood Ward -43.5
## 5 1.21e18 Eco Spa~ 1.79e8 Peng      Christchurch City Riccarton Wa~ -43.5
## 6 1.45e18 Merival~ 1.54e8 Alisa Ka~ Christchurch City Fendalton Wa~ -43.5
## 7 5.27e 7 Studio ~ 4.26e8 Raj       Christchurch City Riccarton Wa~ -43.5
## 8 8.10e17 City li~ 4.97e8 Michael   Christchurch City Heathcote Wa~ -43.6
## 9 3.94e 7 Wigram ~ 3.03e8 Hongmei   Christchurch City Halswell Ward -43.6
## 10 1.46e18 Dollar ~ 2.98e8 Tony      Christchurch City Heathcote Wa~ -43.6
```

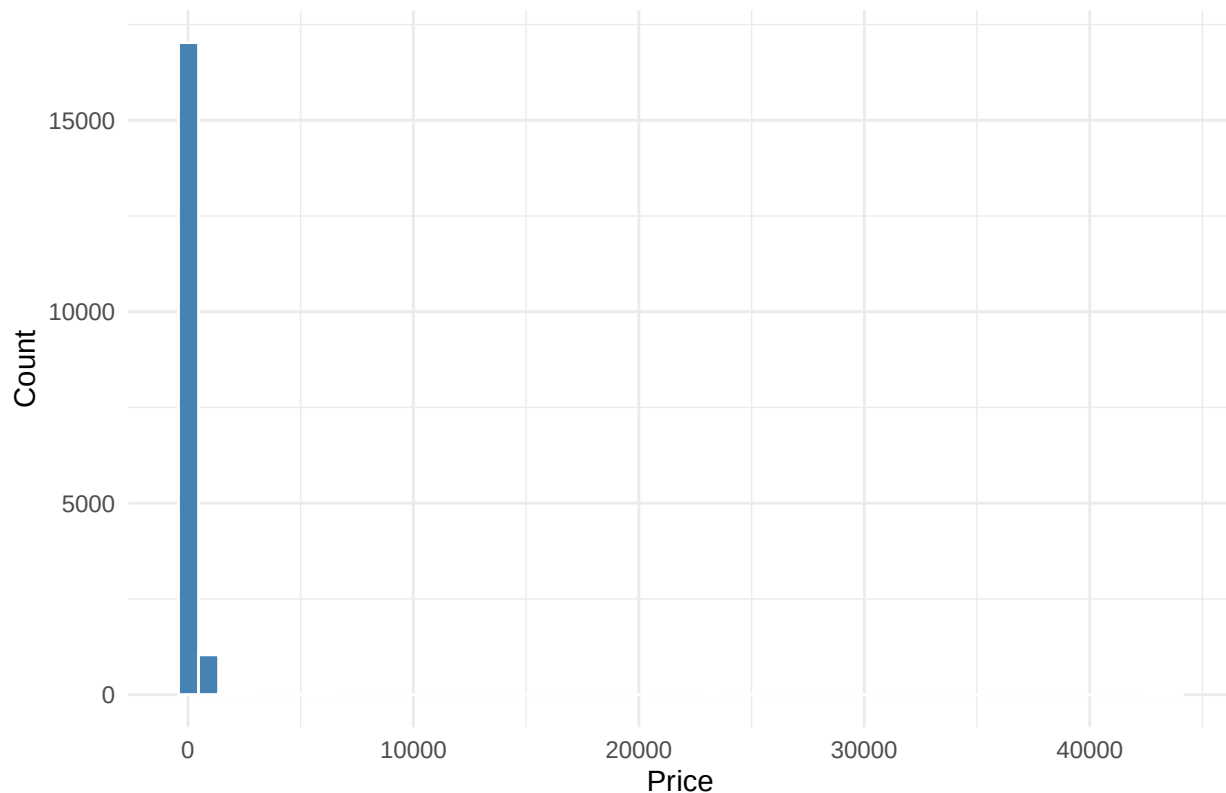
```
## # i 28,785 more rows
```

```
## # i 12 more variables: longitude <dbl>, room_type <chr>, price <dbl>,
## #   minimum_nights <dbl>, number_of_reviews <dbl>, last_review <date>,
## #   reviews_per_month <dbl>, calculated_host_listings_count <dbl>,
## #   availability_365 <dbl>, number_of_reviews_ltm <dbl>, license <lgl>,
## #   month_year <chr>
```

```
##           id          name
##           0            0
##       host_id      host_name
##           0            1
##   neighbourhood_group neighbourhood
##           0            0
##       latitude      longitude
```

```
##          0          0
##          room_type      price
##          0          10667
##          minimum_nights    number_of_reviews
##          37          0
##          last_review      reviews_per_month
##          2627          2627
## calculated_host_listings_count    availability_365
##          0          0
##          number_of_reviews_ltm      license
##          0          28795
##          month_year
##          0
```

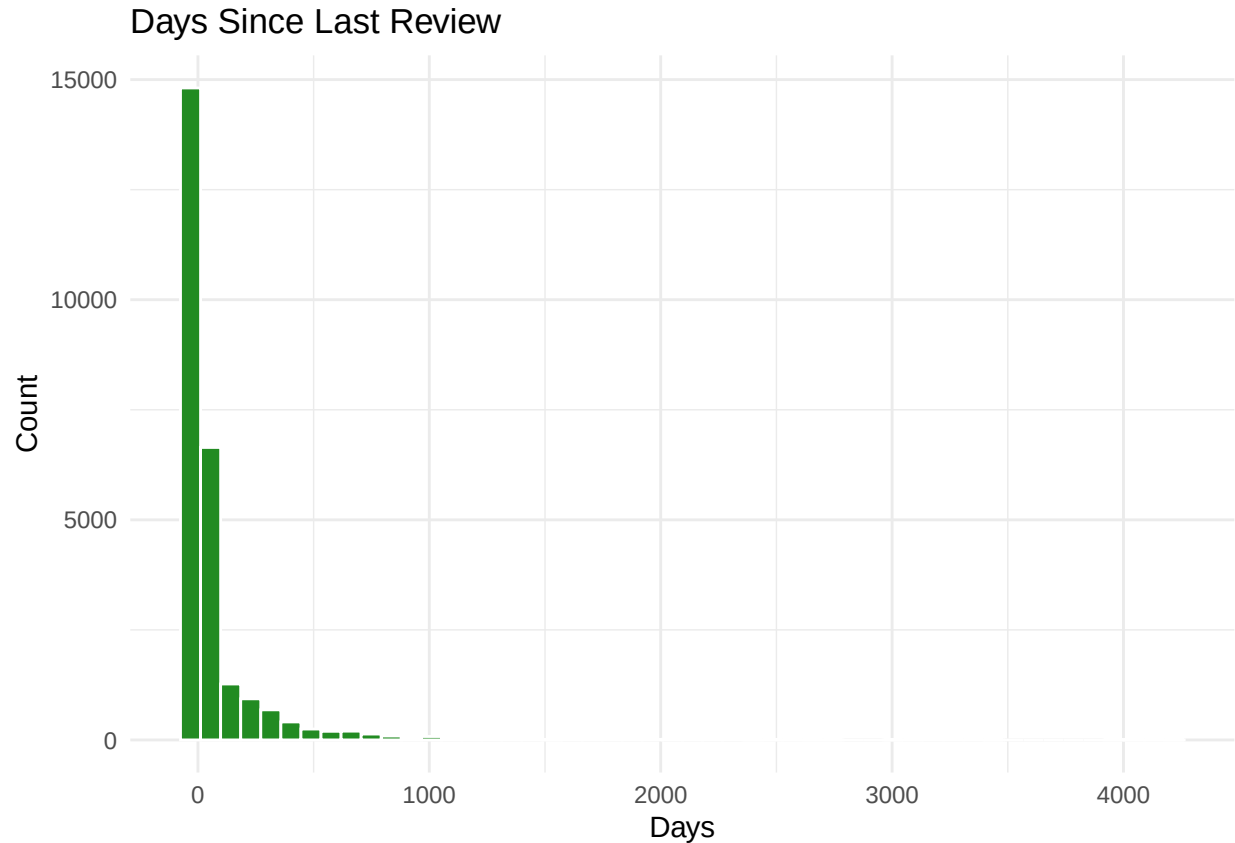
Distribution of Rental Prices



Instruction from week 5 lab - so probably need days\_since\_review column - added that column in

```
## # A tibble: 6 x 21
##       id name      host_id host_name neighbourhood_group neighbourhood latitude
##   <dbl> <chr>      <dbl> <chr>      <chr>      <chr>      <dbl>
## 1 4.02e 7 Ensuite ~ 9.88e7 North So~ Christchurch City Harewood Ward -43.5
## 2 1.15e18 Handy ci~ 2.98e8 Tony      Christchurch City Central Ward -43.5
## 3 1.10e 6 Heritage~ 4.99e6 Paulette   Christchurch City Fendalton Wa~ -43.5
## 4 1.34e18 Settle i~ 5.43e7 Sarena     Christchurch City Burwood Ward -43.5
## 5 1.21e18 Eco Spa~~ 1.79e8 Peng       Christchurch City Riccarton Wa~ -43.5
## 6 1.45e18 Merivale~ 1.54e8 Alisa Ka~ Christchurch City Fendalton Wa~ -43.5
## # i 14 more variables: longitude <dbl>, room_type <chr>, price <dbl>,
```

```
## # minimum_nights <dbl>, number_of_reviews <dbl>, last_review <date>,
## # reviews_per_month <dbl>, calculated_host_listings_count <dbl>,
## # availability_365 <dbl>, number_of_reviews_ltm <dbl>, license <lgl>,
## # month_year <chr>, scrape_date <date>, days_since_review <dbl>
```



```
## # A tibble: 9,418 x 21
##       id name      host_id host_name neighbourhood_group neighbourhood latitude
##       <dbl> <chr>      <dbl> <chr>      <chr>          <chr>          <dbl>
## 1 1.50e18 The Col~ 1.64e8 Joel      Christchurch City Linwood Ward    -43.5
## 2 1.30e18 Tranqui~ 4.96e7 Xihang     Christchurch City Spreydon Ward   -43.6
## 3 8.41e17 Stunnin~ 2.34e7 Hennie     Christchurch City Harewood Ward   -43.5
## 4 4.15e 7 The Lit~ 1.50e8 Gareth    Christchurch City Papanui Ward    -43.5
## 5 1.42e18 Home in~ 9.47e7 Andrew H~ Christchurch City Waimairi Ward   -43.5
## 6 1.06e18 Style m~ 5.51e8 Paula Ka~ Christchurch City Central Ward     -43.5
## 7 1.17e18 Entire ~ 6.83e8 Devon     Christchurch City Fendalton Wa~   -43.5
## 8 1.46e18 Warmth ~ 5.00e8 Anupriya   Christchurch City Innes Ward       -43.5
## 9 4.00e 7 Modern ~ 2.05e7 Christop~ Christchurch City Coastal Ward     -43.5
## 10 3.43e 7 Art, Es~ 6.74e7 Angela     Christchurch City Central Ward     -43.5
## # i 9,408 more rows
## # i 14 more variables: longitude <dbl>, room_type <chr>, price <dbl>,
## # minimum_nights <dbl>, number_of_reviews <dbl>, last_review <date>,
## # reviews_per_month <dbl>, calculated_host_listings_count <dbl>,
## # availability_365 <dbl>, number_of_reviews_ltm <dbl>, license <lgl>,
## # month_year <chr>, scrape_date <date>, days_since_review <dbl>
```

Summary statistics: price: median price is \$205, but the max is \$43,654 (a massive extreme outlier? error?).

Price histogram: right skewed, due to extreme price outliers

license: entirely NA (28,795 missing values) -> consider dropping if doesn't need this column

Missing values: price has 10,667 missing values: how to handle? or filter them out?

last\_review and reviews\_per\_month both have 2,627 missing values

days\_since\_review has 9418 negative values. Days since last review histogram: Most reviews happened very recently (clumped near 0 days), which makes sense for active listings, but there are long-tail stale listings stretching out past 1,000+ days. For month\_year column, change values to include the day the scrape date was done (from the website).

```
## # A tibble: 1,745 x 21
##       id name      host_id host_name neighbourhood_group neighbourhood latitude
##   <dbl> <chr>      <dbl> <chr>      <chr>              <chr>      <dbl>
## 1 4.15e 7 The Lit~ 1.50e8 Gareth   Christchurch City Papanui Ward    -43.5
## 2 1.46e18 Warmth ~ 5.00e8 Anupriya Christchurch City Innes Ward      -43.5
## 3 4.00e 7 Modern ~ 2.05e7 Christop~ Christchurch City Coastal Ward    -43.5
## 4 2.16e 7 Discove~ 1.14e8 Ollie     Christchurch City Heathcote Wa~ -43.6
## 5 1.15e18 Central~ 4.07e7 Sarah    Christchurch City Central Ward   -43.5
## 6 5.16e 7 Bedroom~ 8.17e6 Alison  Christchurch City Riccarton Wa~ -43.5
## 7 9.25e17 Midcent~ 1.05e7 Zia      Christchurch City Central Ward   -43.5
## 8 1.20e18 Black D~ 9.11e7 Holly    Christchurch City Banks Penins~ -43.6
## 9 1.14e18 Bernie'~ 1.97e8 Olivia   Christchurch City Halswell Ward -43.6
## 10 2.02e 7 Hawthor~ 1.43e8 Sarah    Christchurch City Waimairi Ward -43.5
## # i 1,735 more rows
## # i 14 more variables: longitude <dbl>, room_type <chr>, price <dbl>,
## #   minimum_nights <dbl>, number_of_reviews <dbl>, last_review <date>,
## #   reviews_per_month <dbl>, calculated_host_listings_count <dbl>,
## #   availability_365 <dbl>, number_of_reviews_ltm <dbl>, license <lgl>,
## #   month_year <chr>, scrape_date <date>, days_since_review <dbl>
```

Even after matching the scrape dates with the exact publication times, 1745 negative values remain in days\_since\_review column. (So these reviews were posted in the days right after each monthly data collection, counted as active listing. So leave these negative days for now)

DEAL with price column first -> extreme outliers + 10667 missing values Check Price inconsistencies

```
## [1] "numeric"
```

```
## [1] 10667
```

See if 10667 missing prices are random or linked to specific months

```
## # A tibble: 9 x 4
##   month_year total_listings missing_prices pct_missing
##   <chr>          <int>          <int>      <dbl>
## 1 2025-10           2991             170      5.68
## 2 2025-11           3047             212      6.96
## 3 2025-12           3180            3180     100
## 4 2026-01           3057            3057     100
## 5 2026-02           3190            3190     100
## 6 2026-03           3227             168      5.21
## 7 2026-04           3273             204      6.23
## 8 2026-05           3361             212      6.31
## 9 2026-06           3469             274      7.90
```

100% price missing values in 2025-12, 2026-01, 2026-02 months

checking whether a listing has only one unique price across all its available months. If it does, it's stable, can use that price to fill in December, January, and February.

```
##
## FALSE TRUE
## 3535 582
```

FALSE (3,535 listings): For the vast majority of properties, the host never changed the listing title across the entire time period you observed. Their title stayed 100% identical.

TRUE (582 listings): For 582 properties, the host did change their listing title at least once during your dataset

```
##
## 1 2 3 4 5 6
## 547 752 666 696 677 616
```

```
## [1] 10158
```

So 509 prices were filled as they were the same across every single other months they appeared in (id)

Still has 10158 missing prices

Manage price outliers/extreme values Inspect the extreme rows

```
## # A tibble: 20 x 5
##       id name                                price room_type minimum_nights
##       <dbl> <chr>                                <dbl> <chr>          <dbl>
## 1 9.15e17 La Maison de Manu Waiata - Akaroa Hol~ 43654 Entire h~      1
## 2 5.94e17 Seaview Outlook - Wainui Holiday Home 43654 Entire h~      2
## 3 1.00e18 Pacific Panorama - Christchurch Holid~ 43654 Entire h~      2
## 4 1.41e18 Central Cutie - Christchurch City Cot~ 43654 Entire h~      2
## 5 5.94e17 Seaview Outlook - Wainui Holiday Home 43188 Entire h~      2
## 6 9.15e17 La Maison de Manu Waiata - Akaroa Hol~ 43188 Entire h~      1
## 7 1.00e18 Pacific Panorama - Christchurch Holid~ 43188 Entire h~      2
## 8 1.41e18 Central Cutie - Christchurch City Cot~ 43178 Entire h~      2
## 9 1.29e18 Groups wanted CHCH CBD 3bed 3 bath wi~ 22971 Entire h~      1
## 10 1.85e 7 The East Penthouse                    6277 Private ~      1
## 11 1.85e 7 The East Penthouse                    6277 Private ~      1
## 12 1.85e 7 The East Penthouse                    6277 Private ~      1
## 13 1.85e 7 The East Penthouse                    6277 Private ~      1
## 14 1.85e 7 The East Penthouse                    5000 Private ~      1
## 15 1.85e 7 The East Penthouse                    5000 Private ~      1
## 16 1.50e18 Elegant Mansion for Weddings, Busines~ 3029 Entire h~      2
## 17 1.50e18 Elegant Mansion for Weddings, Busines~ 3029 Entire h~      2
## 18 7.92e17 Exquisite Contemporary Haven · Luxury~ 2470 Entire h~      2
## 19 1.50e18 Elegant Mansion for Weddings, Busines~ 2350 Entire h~      2
## 20 1.30e18 Executive House - Huntsbury Hill      2250 Entire h~      5
```

Rows 1-8 -> likely error (checking latitude/altitude), doesn't make sense to have that much a night Rows 10-15 -> also likely error. priced >5000 per night in a private room doesn't make sense Rows 16-20.. maybe? entire home

Keep rows 1-15 will cause statistical bias into analysis. It distorts the mean and standard deviation DECISION: Filter these 15 rows (note: filter() keeps every row where the condition evaluates to TRUE, and throws away anything that evaluates to FALSE)

```
## [1] 15
```

```
## [1] 3029
```

```
## # A tibble: 1 x 4
##   mean_price median_price min_price max_price
##   <dbl>         <dbl>    <dbl>    <dbl>
## 1      238.         204        27      3029
```

What to do with remaining PRICE NAs? - leave them as NAs for now as not sure what is happening downward

License 100% NAs - no universal registration rules so hosts didn't have to register for official registration code to list their property -> DECISION DROP license column

```
## [1] "id"                "name"
## [3] "host_id"           "host_name"
## [5] "neighbourhood_group" "neighbourhood"
## [7] "latitude"          "longitude"
## [9] "room_type"         "price"
## [11] "minimum_nights"    "number_of_reviews"
## [13] "last_review"       "reviews_per_month"
## [15] "calculated_host_listings_count" "availability_365"
## [17] "number_of_reviews_ltm" "month_year"
## [19] "scrape_date"       "days_since_review"
```

check if reviews\_per\_month and last\_review are related

```
## [1] TRUE
```

if property has 0 reviews, both reviews\_per\_month, and last\_review are missing.

```
## [1] 0
```

```
## # A tibble: 0 x 4
## # i 4 variables: id <dbl>, name <chr>, reviews_per_month <dbl>,
## #   last_review <date>
```

There are zero listings.. every single listing either has a positive review or is missing (NA). So reviews\_per\_month and last\_review are 100% structurally locked

DECISION: takes every missing value (NA) in the reviews\_per\_month column and turns it into a literal 0

```
## [1] 0
```

```
## [1] 2619
```

```
##          id          name
##          0          0
##      host_id      host_name
##          0          1
##      neighbourhood_group      neighbourhood
##          0          0
##      latitude      longitude
##          0          0
##      room_type      price
##          0      10158
##      minimum_nights      number_of_reviews
##          37          0
##      last_review      reviews_per_month
##      2619          0
##      calculated_host_listings_count      availability_365
##          0          0
##      number_of_reviews_ltm      month_year
##          0          0
##      scrape_date      days_since_review
##          0      2619
```

The Review-Related Columns (2,619 NAs) last\_review: 2,619 last\_review\_date: 2,619 days\_since\_review: 2,619 These three columns are locked together Decision to leave them as NA since can't calculate "days since a review" if there has never been a review.

Price column: 10158 NAs

minimum\_nights: 37 NAs

TO DECIDE: MINIMUM\_NIGHTS NAs ?impute with median : min-1, max-365, median-1, mean - 1.788

COLUMNS to drop: neighbourhood\_group only has 1 unique value so can't help compare anything

```
## [1] "Christchurch City" "Christchurch City" "Christchurch City"
## [4] "Christchurch City" "Christchurch City" "Christchurch City"
```

DECISION: DROP neighbourhood\_group column

```
## [1] "id"          "name"
## [3] "host_id"     "host_name"
## [5] "neighbourhood" "latitude"
## [7] "longitude"   "room_type"
## [9] "price"       "minimum_nights"
## [11] "number_of_reviews" "last_review"
## [13] "reviews_per_month" "calculated_host_listings_count"
## [15] "availability_365" "number_of_reviews_ltm"
## [17] "month_year"     "scrape_date"
## [19] "days_since_review"
```

```
## [1] FALSE
```

DECISIONS: Drop: host\_name : irrelevant to analysis, and is personal identifiable information

```
## [1] "id" "name"
## [3] "host_id" "neighbourhood"
## [5] "latitude" "longitude"
## [7] "room_type" "price"
## [9] "minimum_nights" "number_of_reviews"
## [11] "last_review" "reviews_per_month"
## [13] "calculated_host_listings_count" "availability_365"
## [15] "number_of_reviews_ltm" "month_year"
## [17] "scrape_date" "days_since_review"

## # A tibble: 0 x 18
## # i 18 variables: id <dbl>, name <chr>, host_id <dbl>, neighbourhood <chr>,
## # latitude <dbl>, longitude <dbl>, room_type <chr>, price <dbl>,
## # minimum_nights <dbl>, number_of_reviews <dbl>, last_review <date>,
## # reviews_per_month <dbl>, calculated_host_listings_count <dbl>,
## # availability_365 <dbl>, number_of_reviews_ltm <dbl>, month_year <chr>,
## # scrape_date <date>, days_since_review <dbl>
```

Check duplicates

```
## [1] 0
```

```
## [1] 24667
```

```
## # A tibble: 3,898 x 2
##       id      n
##   <dbl> <int>
## 1  204050     2
## 2  319770     9
## 3  737452     9
## 4 1004252     9
## 5 1007247     9
## 6 1097487     9
## 7 1429281     9
## 8 1461771     9
## 9 1798383     9
## 10 1837611     9
## # i 3,888 more rows
```

```
## # A tibble: 2 x 4
##       id month_year price availability_365
##   <dbl> <chr>      <dbl>          <dbl>
## 1 204050 2025-10      95            365
## 2 204050 2025-11      98            364
```

The same listing IDs appear multiple times because Inside Airbnb scraped them across different months, capturing how their prices and availability dynamically changed over time.

This means they are not accidental duplicate errors. They are time-series records.

```
## Rows: 28,780
## Columns: 18
```

```
## $ id <dbl> 4.018377e+07, 1.149729e+18, 1.097487e+0~
## $ name <chr> "Ensuite Cabin (Sleeps 6)", "Handy city~
## $ host_id <dbl> 98758734, 298456605, 4993285, 54299306,~
## $ neighbourhood <chr> "Harewood Ward", "Central Ward", "Fenda~
## $ latitude <dbl> -43.47002, -43.53270, -43.51149, -43.48~
## $ longitude <dbl> 172.5709, 172.6640, 172.6254, 172.6687,~
## $ room_type <chr> "Private room", "Private room", "Privat~
## $ price <dbl> NA, NA, 166, NA, 126, 585, 145, 450, NA~
## $ minimum_nights <dbl> 1, 1, 1, 2, 1, 2, 1, 2, 1, 1, 7, 1, 1, ~
## $ number_of_reviews <dbl> 0, 25, 230, 2, 8, 0, 6, 0, 1, 0, 1, 5, ~
## $ last_review <date> NA, 2025-07-15, 2025-03-03, 2025-04-26~
## $ reviews_per_month <dbl> 0.00, 1.47, 2.17, 0.35, 0.58, 0.00, 0.2~
## $ calculated_host_listings_count <dbl> 20, 6, 1, 1, 20, 3, 4, 1, 2, 6, 4, 14, ~
## $ availability_365 <dbl> 317, 0, 221, 0, 359, 78, 348, 363, 4, 0~
## $ number_of_reviews_ltm <dbl> 0, 21, 14, 2, 5, 0, 2, 0, 0, 0, 1, 4, 2~
## $ month_year <chr> "2025-10", "2025-10", "2025-10", "2025--~
## $ scrape_date <date> 2025-10-05, 2025-10-05, 2025-10-05, 20~
## $ days_since_review <dbl> NA, 82, 216, 162, 218, NA, 267, NA, 210~
```

```
## [1] 28780
```

Summary:

1. price column Initially had 10,667 missing values, with 100% missing values for the 3 months: Dec 2025–Feb 2026. Imputed 509 prices where the listing had the same price across all other months. Removed 15 extreme outliers with prices > \$5,000, which were likely errors. Before: 28,795 rows After: 28,780 rows Question: What should we do with the remaining 10,158 missing values? The price column is still right-skewed.
2. days\_since\_review column. Added days\_since\_review as instructed in Week 5. Initially, scrape\_date was set to the first day of each month (YYYY-MM-01), resulting in 9,418 negative values. Updated scrape\_date to use the actual day the listing was scraped and recalculated days\_since\_review. There are now 1,745 negative values, likely because some reviews were posted after the scrape date. Question: Do we need this column? If so, how should we handle the negative values?
3. Removed columns: license – removed because it was 100% missing. neighbourhood\_group – removed because all values were Christchurch\_city, so it provides no useful comparison. host\_name – removed because it contains personally identifiable information and is not relevant to the analysis.
4. reviews\_per\_month and last\_review: Converted 2,619 missing values in reviews\_per\_month to 0, since listings with no reviews have both reviews\_per\_month and last\_review recorded as NA. Left last\_review NAs unchanged for now, since replacing them with 0 would not make sense for a date variable.
5. minimum\_nights: Has 37 missing values. Current statistics: min = 1, median = 1, mean = 1.788, max = 365. Have left as it is. Depend on analysis later - impute missing values using median (1) or leave as it is?
6. Other findings: There were no exact duplicates. The same listing IDs appear multiple times because Inside Airbnb scraped them across different months, capturing how their prices and availability changed over time. So they are time-series records.

Before cleaning- number of rows: 28795, columns: 19 After cleaning: 28780, columns: 18

(Note: Cleaned dataset: df\_cleaned)